

CHURCH FENTON, United Kingdom

WMO: 033550

Lat: 53.834N

Lon: 1.196W

Elev: 29

StdP: 14.68

Time Zone: 0.00 (EUW)

Period: 01-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	23.7	26.8	19.9	15.0	29.7	22.5	16.9	30.8	33.0	49.3	29.7	49.9	4.4	330	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.8	77.6	64.9	74.3	63.2	71.3	61.4	66.6	74.3	64.6	71.9	62.9	69.2	9.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.7	88.3	69.6	61.8	82.6	67.5	60.2	78.0	65.8	31.2	74.5	29.6	71.8	28.5	69.3	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
27.6	23.6	20.8	DB	17.3	83.6	5.7	2.9	13.2	85.7	9.8	87.4	6.6	89.0	2.4	91.2
			WB	16.7	72.0	5.7	5.6	12.6	76.0	9.2	79.3	6.0	82.4	1.9	86.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.8	40.3	40.1	43.1	47.7	52.6	58.2	61.7	61.3	57.3	51.1	44.4	39.1	(6)
	DBStd	9.67	6.17	6.25	5.36	5.06	4.96	4.44	4.30	3.87	4.35	5.20	5.85	6.82	(7)
	HDD50	1509	302	280	220	103	29	1	0	0	2	49	181	342	(8)
	HDD65	5588	765	697	678	520	384	208	124	129	232	431	617	803	(9)
	CDD50	1437	2	4	7	33	110	248	362	349	222	83	14	3	(10)
	CDD65	42	0	0	0	0	0	6	21	13	2	0	0	0	(11)
	CDH74	319	0	0	0	0	9	50	169	65	22	4	0	0	(12)
	CDH80	32	0	0	0	0	0	5	14	11	1	1	0	0	(13)
(14) Wind	WSAvg	9.6	11.4	10.3	10.2	9.7	9.9	8.4	8.2	8.8	9.0	9.4	10.0	9.7	(14)
(15) Precipitation	PrecAvg	23.1	1.9	1.5	1.8	1.7	1.9	2.0	1.8	2.4	2.0	1.9	2.0	2.2	(15)
	PrecMax	28.8	3.5	4.6	4.1	3.7	4.8	5.7	4.1	5.1	5.9	4.6	4.3	6.3	(16)
	PrecMin	15.6	0.4	0.0	0.3	0.3	0.4	0.2	0.2	0.4	0.3	0.4	0.6	0.9	(17)
	PrecStd	3.5	0.9	1.0	0.9	0.9	1.1	1.3	1.0	1.2	1.4	1.0	0.9	1.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.2	57.3	61.2	68.8	74.8	80.0	81.8	81.1	76.9	67.2	60.0	56.7	(19)
		MCWB	51.7	51.5	51.6	56.6	63.7	67.2	65.9	69.5	61.7	60.1	55.9	52.2	(20)
	2%	DB	53.9	54.2	57.7	63.9	69.5	74.3	78.6	74.8	71.8	63.3	57.1	53.9	(21)
		MCWB	49.9	50.1	50.0	53.2	58.8	62.5	65.7	64.7	61.4	57.9	53.1	50.3	(22)
	5%	DB	51.7	51.8	55.1	60.6	65.7	71.1	75.2	71.8	68.6	61.1	55.3	51.7	(23)
		MCWB	48.1	47.7	49.0	52.0	56.2	61.0	63.9	62.4	59.8	56.6	51.9	48.7	(24)
	10%	DB	49.7	49.6	52.6	57.9	62.3	67.9	71.0	69.5	65.8	59.2	53.3	49.2	(25)
		MCWB	46.3	45.7	47.2	50.4	54.2	59.2	61.3	61.0	58.1	54.7	50.0	46.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.3	53.1	53.5	58.3	63.3	68.6	69.3	71.7	65.4	62.0	56.8	52.7	(27)
		MCDB	55.4	55.9	58.4	65.1	73.4	77.1	78.0	77.1	72.6	66.5	59.5	55.4	(28)
	2%	WB	50.3	50.4	51.4	55.5	59.8	64.6	66.3	66.5	62.8	59.4	54.1	50.9	(29)
		MCDB	53.2	53.7	56.5	61.7	68.0	71.2	75.4	72.3	69.0	62.7	56.5	53.4	(30)
	5%	WB	48.5	48.3	49.6	53.0	57.4	62.3	64.5	64.2	61.0	57.4	52.2	49.2	(31)
		MCDB	51.1	51.4	54.3	58.9	63.2	68.8	73.4	69.8	66.2	60.3	54.6	51.5	(32)
	10%	WB	46.7	46.4	47.7	51.0	55.4	60.0	62.7	62.5	59.6	55.6	50.4	46.9	(33)
		MCDB	49.4	49.1	52.3	56.8	60.9	66.5	69.6	67.2	64.5	58.4	53.0	49.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	11.0	13.8	16.4	15.9	15.9	15.8	15.0	15.1	12.4	10.9	9.8	(35)
		MCDBR	11.5	12.6	18.0	21.1	21.7	22.2	22.9	19.0	19.7	14.0	11.8	10.9	(36)
	5% WB	MCWBR	9.9	9.8	11.4	12.3	12.3	12.3	11.2	10.0	10.9	9.5	9.8	9.8	(37)
		MCWBR	11.6	11.8	16.5	19.8	20.1	20.0	21.6	16.2	17.1	12.9	11.6	10.8	(38)
(40) Clear-Sky Solar Irradiance	taub		0.321	0.337	0.368	0.396	0.395	0.400	0.401	0.388	0.368	0.349	0.323	0.309	(40)
		taud	2.342	2.385	2.301	2.237	2.278	2.290	2.293	2.349	2.413	2.442	2.398	2.318	(41)
	Ebn at Noon		202	240	255	263	270	269	266	264	254	233	201	180	(42)
		Edh at Noon	18	24	32	39	39	39	38	34	29	23	17	16	(43)
(44) All-Sky Solar Radiation	RadAvg		204	413	755	1179	1491	1531	1486	1225	897	507	251	161	(44)
	RadStd		22	47	94	138	93	140	146	80	82	44	27	16	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (64 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page